

Object Following Robot using Arduino Uno



MINI-PROJECT PRESENTATION

Department of Electronics and Communication Engineering

Cooch Behar Government Engineering College

Maulana Abul Kalam Azad University of Technology (MAKAUT)

PROJECT MEMBERS

Bikkram Das (34900323012) | Biraj Sarkar (34900323013) | Debanjan Chakraborty (34900323015)
| Sayan Sarkar (34900323048) | Soumyajit Pandit (34900323052)

PROJECT SUPERVISOR Dr. Gautam Das

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Introduction & Project Overview

Autonomous target following

Following is a core capability of Human–Machine Interaction (HMI) — enabling hands-free operation and natural cooperation between people and mobile platforms.

Why not a camera?

Vision-based tracking (e.g. Raspberry Pi + camera) needs heavy processing, more weight, cost and power. Acoustic and IR sensors are a low-cost, low-power alternative.

Multi-sensor configuration

A front HC-SR04 ultrasonic sensor plus two side-mounted IR sensors together define a clean spatial envelope for real-time following.

Real-world applications

- Industrial automation carts
- Hospital carts following staff
- Hands-free warehouse transport
- Automated shopping-guide robots

Design goal:

low-cost | low-power | real-time

System Architecture

Sensor input

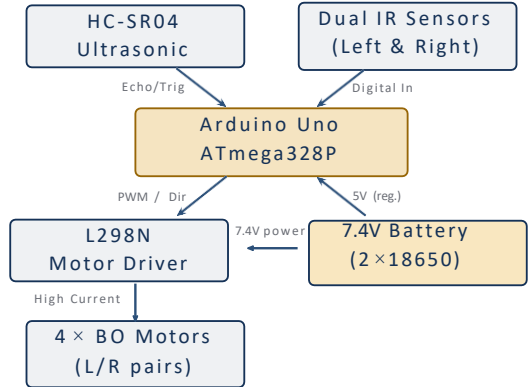
HC-SR04 gives front distance; left/right IR modules flag lateral offset.

Central processing

Arduino Uno runs a state machine that turns sensor readings into direction and PWM speed commands.

Actuation

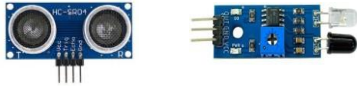
L298N H-bridge drives four BO geared motors; steering is by differential wheel speed.



Hardware: Arduino Uno & Sensors



Arduino Uno — ATmega328P, 16MHz, 14 digital I/O
(6 PWM), 6 analog in



HC-SR04 ultrasonic sensor (left) and IR proximity
module (right)

Arduino Uno board

Runs the control loop that reads every sensor and regulates the actuators each cycle.

HC-SR04 ultrasonic sensor

Emits 40 kHz pulses; distance from echo time:

$$d = \frac{t \times 0.034}{2} \text{ cm}$$

Range 2–400 cm, accuracy ± 3 mm.

IR proximity sensors

Dual side-mounted units emit IR light; a reflection lowers photodiode resistance, driving the output LOW when a target is within 8 cm.

Actuators & Motor Driver

L298N H-bridge driver

Switches high-power transistors to control DC-motor speed and direction; up to 2 A per channel. Average motor voltage is set by PWM on the Enable pins (ENA, ENB).

Brushed DC BO geared motors

Four motors, 3–9 V operating range, 1:48 gear reduction → 150 RPM at 6 V with high torque.

4WD kinematics

Motors wired as parallel left/right pairs. Differential speed between the pairs gives forward drive, reverse, and pivot turns.



L298N H-bridge module



BO geared DC motor

Pin Configuration

ARDUINO UNO – PIN CONFIGURATION

1. HC-SR04 ULTRASONIC SENSOR

HC-SR04 Pin	Arduino UNO Pin
VCC	5V
GND	GND
TRIG	D9
ECHO	D10

2. LEFT IR SENSOR

IR Sensor Pin	Arduino UNO Pin
VCC	5V
GND	GND
OUT	D11

3. RIGHT IR SENSOR

IR Sensor Pin	Arduino UNO Pin
VCC	5V
GND	GND
OUT	D12

4. L298N MOTOR DRIVER

L298N Pin	Arduino UNO Pin
ENA	D5 (PWM)
IN1	D6
IN2	D7
IN3	D8
IN4	D4
ENB	D3 (PWM)
GND	GND (Common Ground)
5V	5V (from Arduino or L298N 5V OUT)

5. MOTOR CONNECTIONS (L298N OUTPUT)

L298N Output	Connected To
OUT1	Left Motor 1 (Front) +
OUT2	Left Motor 2 (Rear) + (Parallel with Left Motor 1)
OUT3	Right Motor 1 (Front) +
OUT4	Right Motor 2 (Rear) + (Parallel with Right Motor 1)

6. POWER CONNECTIONS

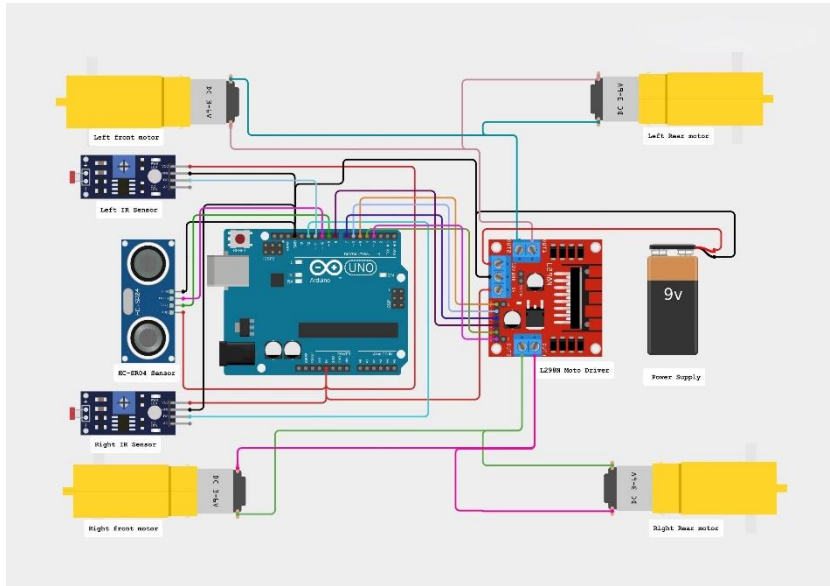
Power Source	Connect To
Battery (+)	L298N 12V / VIN
Battery (-)	L298N GND
Arduino GND	L298N GND (Common Ground)
Arduino 5V	Sensors VCC (or L298N 5V OUT if used)

7. ARDUINO UNO DIGITAL PIN SUMMARY

Arduino Pin	Connected Device
D3	L298N ENB (PWM)
D4	L298N IN4
D5	L298N ENA (PWM)
D6	L298N IN1
D7	L298N IN2
D8	L298N IN3
D9	HC-SR04 TRIG
D10	HC-SR04 ECHO
D11	Left IR Sensor OUT
D12	Right IR Sensor OUT

NOTE: All grounds (Arduino GND, L298N GND, Battery -) must be connected together (common ground).

Circuit Diagram



Power Scheme & System Stability

The problem

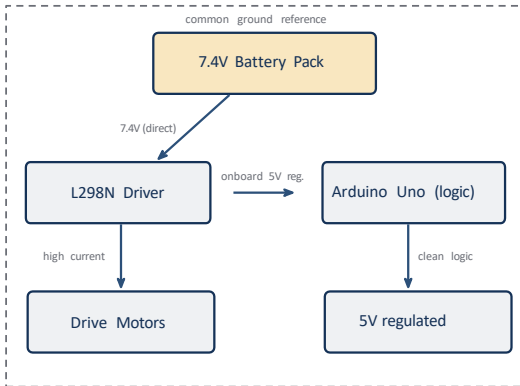
High DC-motor starting current pulls battery voltage down, resetting the microcontroller. An earlier prototype (with a servo + OLED) browned out during active scans.

The fix: split power rails

The 7.4V pack feeds the L298N directly; the driver's onboard 5V regulator powers the Arduino separately, so motor current spikes never touch the logic rail. All grounds are tied together for a common reference.

Outcome

Servo and OLED were dropped from the final build; the rail split alone gave stable, continuous operation.



Isolation prevents motor-current spikes from resetting the MCU.

Software Control Algorithm

Control framework

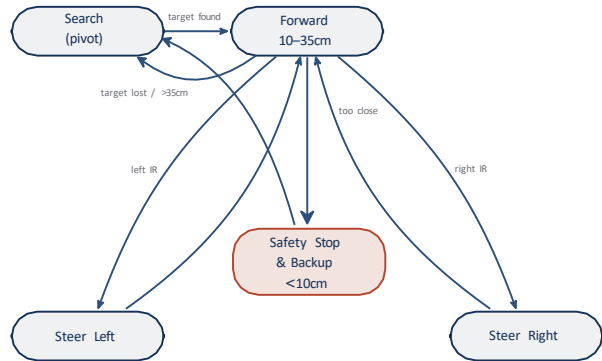
Written in C++ (Arduino IDE 2.x). The NewPing library gives non-blocking acoustic ranging while digital pins are polled for IR state.

Closed-loop transitions

Distance and lateral IR states are continually re-evaluated, triggering one of five direction codes each cycle.

Speed control

Follow envelope: 10–35 cm. PWM is modulated (170/220) based on distance to slow down as the target gets close.



Control Firmware — Source Code

Core `loop()` structure

Reads ultrasonic distance
and both IR flanks each cycle
to pick a motor action.

Safety backup & turn

Target closer than 10 cm →
halt, reverse, turn right, halt
— then re-evaluate.

Rotating search

Target lost or beyond 35 cm
→ robot pivots right to
re-acquire.

```
1 void loop() {  
2   int distance = sonar.ping_cm();  
3   if (distance == 0) distance = 200;  
4   int rightIR = digitalRead(RIGHT_IR);  
5   int leftIR  = digitalRead(LEFT_IR);  
6   analogWrite(ENA, (distance > 25) ? 220 : 170);  
7   analogWrite(ENB, (distance > 25) ? 220 : 170);  
8  
9   if (distance < MIN_DISTANCE) {           // Safety Stop & Backup  
10    setMotors('S'); delay(300);  
11    setMotors('B'); delay(400);  
12    setMotors('S'); delay(200);  
13    setMotors('R'); delay(300);  
14    setMotors('S'); return;  
15  }  
16  if (rightIR == LOW && leftIR == HIGH) setMotors('R');  
17  else if (rightIR == HIGH && leftIR == LOW) setMotors('L');  
18  else if (distance <= FOLLOW_DISTANCE) setMotors('F');  
19  else setMotors('R');                  // Search Mode  
20 }
```

Sensor threshold configuration

IR sensors are trimmed via onboard potentiometers to ignore ambient light and trigger reliably at a maximum range of 8 cm.

Physical integration

All modules mount on a 4-wheel acrylic chassis; a common ground plane removes electrical noise and signal fluctuation.



Command-state calibration map

Condition	Action
Distance < 10 cm	Safety backup & turn
10 cm – 35 cm	Drive forward
Left IR triggered	Pivot left
Right IR triggered	Pivot right
Distance > 35 cm	Target lost → search

Performance Testing & Evaluation

± 0.5 cm

ranging variance

10 ms

control loop latency

50 min

continuous runtime

Target tracking

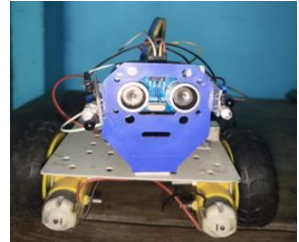
Maintains the 10–35 cm follow boundary reliably, no manual input.

Ranging & latency

10 ms loop latency enables real-time reaction to target speed changes.

Power runtime

Dual 18650 cells sustain full motor load for 50 continuous minutes.



Conclusions

Successful prototype

Built a low-cost, low-power autonomous robot that tracks and follows a target in real time, validating the core design.

Verified differential steering

Differential drive combined with state-command sensor feedback handles lateral and forward movement cleanly.

Coordinated sensor control

Simple acoustic ranging plus IR proximity gives a reliable tracking envelope — no expensive vision hardware required.

Practical experience

Hands-on exposure to hardware–software co-design, power isolation, sensor calibration, and closed-loop control.

A complete, working object-following platform built entirely on low-cost, readily available components.

Active scanning servo

Re-integrate an SG90 micro servo to sweep the ultrasonic sensor across a 180° field of view.

Onboard OLED display

Re-introduce a 0.96" SSD1306 screen to show live distance and operating status.

LM2596 power isolation

Add a step-down buck converter to isolate servo current spikes from the logic rail, removing the earlier reset issue for good.

ESP32-CAM recognition

Upgrade the controller to an ESP32 with onboard camera for face detection and human recognition.

PID control steering

Add a Proportional–Integral–Derivative loop for smoother acceleration and turning, replacing today's fixed-speed steps.

Thank You

Questions & Discussion